



**The finches** on the Galapagos Islands were key to Darwin's work. The 13 species he found there all had different types of beaks, which had evolved to deal with the food available to the birds.

## Natural selection

At the heart of Darwin's idea was his contention that all animal life was derived from a single, common ancestor—that the ancestors of all mammals, humans included, for example, were fish. And in a natural world that was never less than relentlessly violent, only those most able to adapt would survive, in the process evolving into new species.

These views were largely formed by the around-the-world voyage he made as the naturalist on the British survey vessel HMS *Beagle* between 1831 and 1836, most of it spent in South America. It would take him 10 years to work up his voluminous notes and to go through all the samples he collected on his voyage.

Darwin's book inevitably generated controversy, outraging Christian views that the world had been created intact

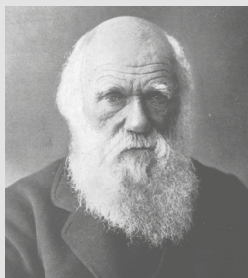
and unchanging by a benevolent deity. Yet however heated the initial debate, quite rapidly there was widespread acceptance of Darwin's views and a realization that he had made a decisive contribution to the understanding of the world. In the process, the status of science generally was immensely boosted.

## The primacy of science

Despite everything, it was possible for Darwinism to be warped. What came to be called "the survival of the fittest" would later prove to be influential in justifying imperialism, racism, and eugenics.

*On the Origin of Species* was published at a time when a growing understanding of the natural world and rapid technological progress meant scientific study had a greater practical worth than ever before. Darwin was one of the last amateur gentleman scientists in a discipline that was becoming professionalized as society came to view science more highly. Partly as a result of Darwin's work, but also because of these changing attitudes, science began to have a more central place in public life. By the end of Darwin's life, continual progress in scientific knowledge had become an almost standard expectation. ■

## Charles Darwin



Charles Darwin (1809–1882) was only the fifth choice for the position of naturalist on the voyage of the HMS

*Beagle* in 1831. However fortuitous his selection, it would transform his life. Although he was constantly seasick during his time aboard the craft, Darwin proved himself an assiduous observer of the world around him. He would take as much amazed delight in the jungles of Brazil as he would in the pampas of Argentina or in the arid wastes of the Galapagos Islands. Upon returning to England, he settled into a life of persistent hard work—the model high-Victorian scientist,

aided by considerable private means and a notably happy family life, despite the deaths of three of his ten children. Although his own health may effectively have been severely damaged by the time he spent on the *Beagle*, his output remained prodigious, as did his level of intrigue at almost any subject in the natural world. In the absence of the exotic, he was as fascinated by pigeons as by parasites, barnacles as much as earthworms.